Assignment 1

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Introduction

This report investigates whether higher marketing spend per customer leads to increased purchase amounts in an online retail setting. The dataset represents a personalised marketing campaign where spending varied across customers through channels such as advertisements, emails, and coupons. Businesses often face uncertainty in determining how much to invest in marketing to maximise returns without overspending. The findings will help determine whether increased spending generates meaningful returns or leads to diminishing benefits.

Research Question

This report examines whether higher marketing spend per customer leads to greater purchase amounts. Specifically, it investigates the relationship between the amount spent on personalised marketing efforts and the revenue generated from each customer, motivated by the need for businesses to optimise marketing investment decisions. The analysis aims to determine whether increased spending results in higher returns or diminishing effectiveness.

Dataset Introduction

The dataset records customer-level information from a personalised marketing campaign, including marketing spend, purchase amount, and number of visits. The key variables of interest are marketing_spend, which represents the amount spent targeting each customer, and purchase_amount, which reflects the revenue generated from each customer. Additional variables such as visits, customer segment, and signup date provide further context on customer activity and characteristics, although they are less central to the research question. Before analysis, the dataset was cleaned by removing rows with missing values in marketing_spend and purchase_amount, as these variables are essential to the investigation. Variable formats were standardised by converting marketing spend, purchase amount, and visits to numeric types, signup_date to a date, segment to a factor, and removing the notes variable as it was not relevant to the analysis.

Table 1: Summary statistics by customer segment

	segment	average_spend	average_purchase	average_visits	number_of_customers
1	Consumer	156.46	147.11	5.51	76
2	Enterprise	151.38	136.48	4.94	16
3	SMB	145.16	147.19	5.15	40

Dataset Description

The dataset contains **132** observations and **6** variables, capturing customer-level information from a personalised marketing campaign. It includes a mix of numeric variables (e.g., marketing_spend, purchase_amount, visits), a categorical variable (segment), and a date variable (signup_date). The segment variable categorises customers into groups such as Consumer, Enterprise, and SMB, enabling analysis of differences in marketing effectiveness and purchasing behaviour across customer types. Marketing spend ranges from **\$3.71** to **\$299.13**, while purchase amounts range from **\$0** to **\$351.87**. Customer visits range from **1** to **12**.

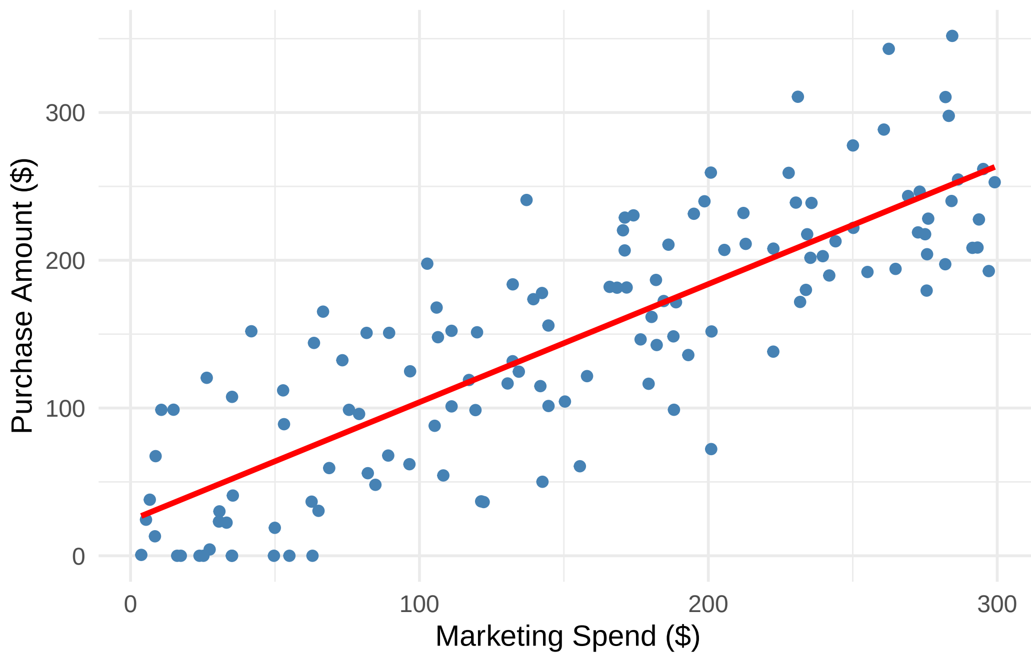

Figure 1: Correlation between marketing spend and purchase amount

Results Section

The results indicate that higher marketing spend is associated with higher purchase amounts across all customer segments. As shown in Figure 2, there is a clear positive relationship between marketing spend and purchase amount, with an overall correlation of **0.83**. While the overall direction of the relationship is consistent, there is noticeable variation in the spread of data points, particularly at higher levels of marketing spend, suggesting some differences in purchasing behaviour within segments.

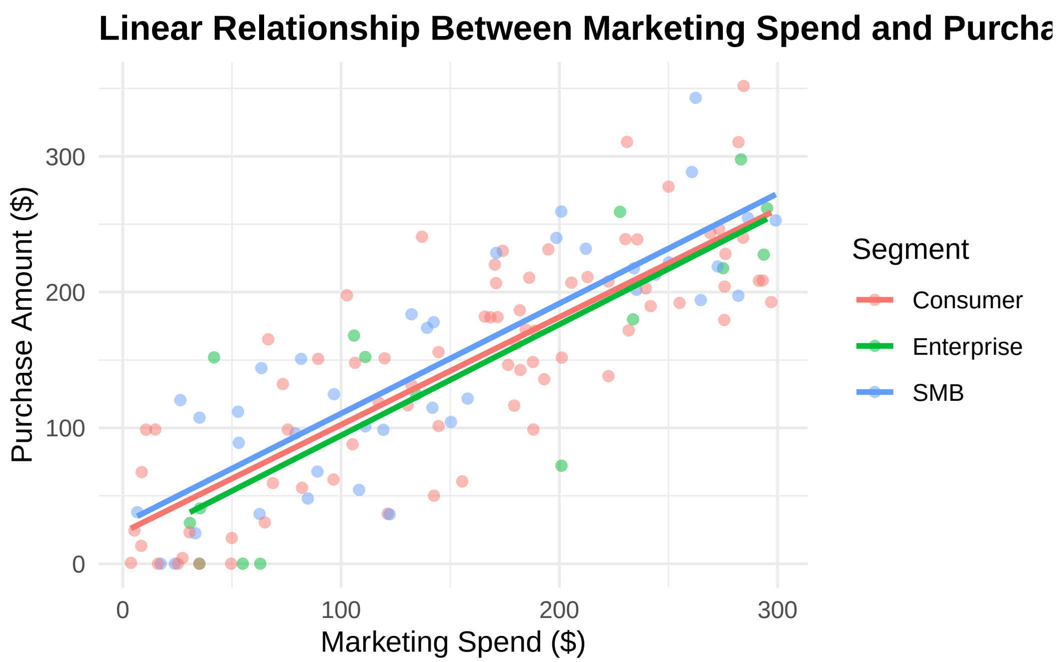

Figure 2: Linear Relationship Between Marketing Spend and Purchase Amount by Customer Segment

Figure 3 provides a clearer view of the underlying trends by smoothing the relationship between marketing spend and purchase amount for each segment. The Consumer and SMB segments display relatively steady increases in purchase amount as marketing spend rises, with average purchase amounts of **$147.11** and **$147.19**, respectively. In contrast, the Enterprise segment shows a less consistent pattern, with a flatter trend at mid-level spend and a lower average purchase amount of **$136.48**, before increasing more sharply at higher levels.

Overall, these findings suggest that while increased marketing spend generally leads to higher purchase amounts, the strength and consistency of this relationship vary across segments. This indicates that the effectiveness of marketing investment may differ depending on customer type, but overall supports the conclusion that higher marketing spend is associated with increased customer purchases.

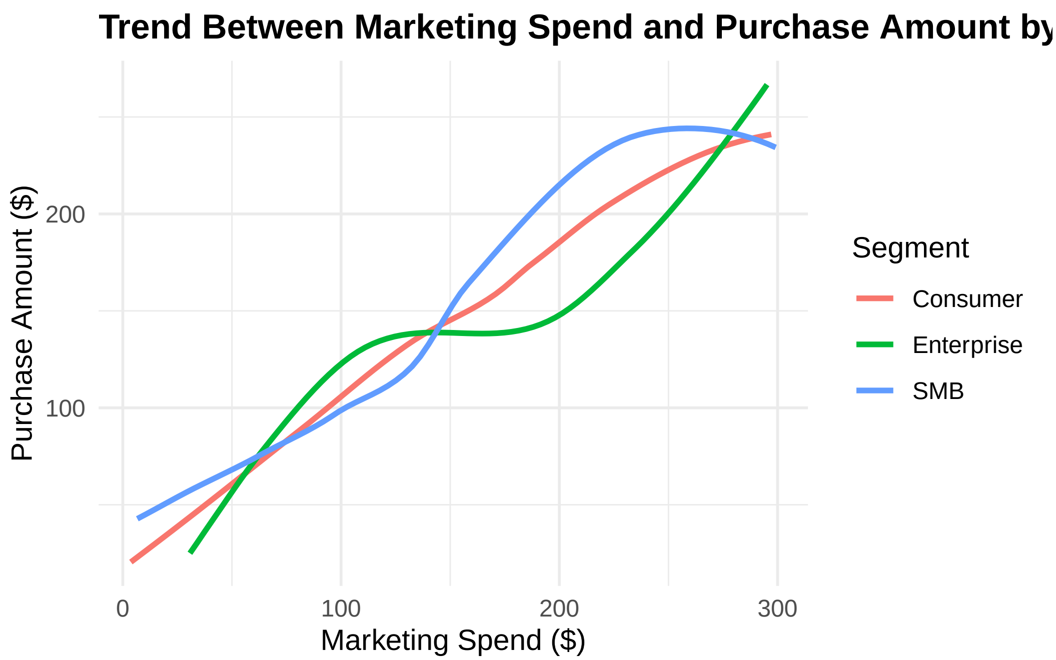

Figure 3: Varying Response of Purchase Amount to Marketing Spend Across Customer Segments